

# LARRIS XIE

larris.xie@uwaterloo.ca · 437-974-6166 · larrisx.com [↗](#) · LinkedIn [↗](#) · GitHub [↗](#) · Scholar [↗](#)

## EDUCATION

University of Waterloo  
Bachelor's of Computer Science

Apr 2028  
3.9/4.0 GPA

## EXPERIENCE

### Conway

Apr 2026 - Aug 2026

*Software Engineering Intern*

- Selected as 1 of 40 **Kleiner Perkins Fellows** from 5000+ candidates
- Created an **durable agent orchestration runtime** for fraud response using the **Claude Agent SDK**, **Python**, and **DBOS**, executing workflows over **autoscaling K8s Jobs**.
- Architected a live **SQL rules engine** in Python and Snowflake that compiles stateful SQL into Snowflake Tasks, powering fraud detection across 1.7B+ events at peak volumes of **17M events/day**.
- Built a **sandboxed browser agent** on **AWS ECS and Kubernetes**, using warm runner pools and pinned NAT Elastic IPs to cut browser session startup by **60x (2 min to under 2s)**.

### University of Waterloo

Jan 2025 - May 2026

*Undergraduate Research Assistant — Security and Privacy in ML*

- **Published as a co-first author** [↗](#) on PPHH-VFL, a **privacy preserving inference architecture** for transformer-based vertical federated learning on distributed time-series data.
- Developed custom **PyTorch transformer** models with **adversarial GAN privacy training**, achieving a 95% F1 score on fraud detection under a 10,000:1 fraud-to-legitimate transaction ratio.
- Accelerated **private transformer inference** by **up to 44.4×** while retaining 100% of performance by engineering a distributed GPU framework with CrypTen MPC and PyTorch RPC.

### Shopify

May 2025 - Aug 2025

*Software Engineering Intern*

- Orchestrated a **100% uptime data migration** from legacy APIs to new service endpoints while ensuring full **backwards compatibility** through shadow writes, backfilling **over 5B rows**, and staged rollouts.
- Designed and implemented scalable **data models and GraphQL APIs** in Ruby on Rails, powering a fully extensible cash management experience for **2M+ active retailers**.
- Engineered full-stack features for Shopify Point of Sale (POS) in React Native and Typescript, resolving critical race conditions in async workflows to reduce failure rate by **40%**.

### Meta x MLH

May 2025 - Aug 2025

*Production Engineering Fellow*

- Deployed and managed **Linux infrastructure** on CentOS, leveraging system calls and kernel-level debugging at the OS layer, and Docker and NGINX at the application layer to improve reliability.
- Configured system monitoring with **Prometheus, cAdvisor, and Grafana**, leveraging low-level resource metrics (CPU, cache misses, memory usage, I/O throughput) to cut bug detection time by **80%**.
- Automated **CI/CD workflows** with GitHub Actions, integrating testing frameworks, security scans, and performance monitoring to streamline deployments and improve system reliability.

## PUBLICATIONS

Fenaux, Lucas\*, **Larris Xie\***, Aditya Bang, et al. (2026). "Private Vertical Federated Inference for Time-Series." *arXiv preprint arXiv:2605.08343*. Under submission at **NeurIPS 2026**. [↗](#) \* Equal contribution.

## PROJECTS

### LifeOS (Hack the 6ix Winner) [↗](#) React, TypeScript, Python, FastAPI, Redis, Docker, PostgreSQL, AWS

- Created a real-time multimodal MCP layer that continuously captures in-person experiences and turns them into searchable, actionable context for LLMs and automations.
- Engineered a **distributed AI video processing pipeline** using Redis Queue that orchestrates semantic analysis with TwelveLabs, embedding generation, and cloud storage via **asynchronous workers**.

### ASCII Game Engine [↗](#) C++

- Architected a 2D game engine in C++ from scratch, grounded in **MVC** architecture and **SOLID principles**, and leveraging **advanced design patterns** for modular and efficient resource management.
- Features **OOP** in console-based rendering, event-driven input and audio, collision detection, dynamic world state with multi-layered rendering and animation, with two demo games: Flappy Bird and Space Invaders.